moving to curves, recursive algorithms (or their iterative counterparts) are needed.

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[1] A knowledge graph is an extremely advanced data structure that organized data in form of graph providing in a single DS the data itself and a way to understand it; Google's knowledge graph, for example, is used to refine their searches through semantic.

[2] Graph databases leverage the fact that modern data is highly interconnected, to allow organizing and querying information in a semantic way, using graph's edges to model dynamic relations between pieces of data (the graph's vertices). This can be thought as taking a relational DB, like a classical SQL DB, and making it more flexible and even more powerful.
Take a look, for instance, at NeoJ in action, or Fullstack GraphQL.

[3] In this case, for PCBs, the edges are limited to polylines made of erpendicular segments

[4] For instance, nobody would be expected to understand Google's knowledge graph by taking a glance at it (especially considering it has a humongous number of vertices and edges): that graph, or graph databases, are not meant to be processed by the human mind, but through algorithms.

[5] A Jordan curve is a plane, simple and closed curve, a non-self-intersecting continuous loop in the plane.

[6] We'll also have to discuss what makes an embedding the best or at least better: "having fewer intersections" is a good starting point, though.

[7] While we present ways to solve this problem, we'll introduce new algorithms and techniques that can also be applied to other areas besides graphs.

[8] It doesn't matter how we label the vertices, because vertices are isomorphic: equivalent one to another, and each adjacent to all the other vertices.

[9] Besides being computationally-heavy, state-of-the-art computer vision needs a large dataset and long time to train, and obviously it doesn't provide a deterministic algorithm.

[10] The first condition can be checked in constant time if we have the information about the size of the graph.

[11] We won't have the space here to describe this algorithm, but you can find a description online, f.i. on Wikipedia https://en.wikipedia.org/wiki/Biconnected_component.

[12] A recurrence relation is an equation that recursively defines a sequence of values, where each term of the sequence is defined as a function of the preceding terms. Check out appendix B for more on how to solve recurrence relations.

[13] Imagine, for the sake of providing a simple example, that at line #2 we are using the empty graph as base, to halt the recursion.